Course notes

Vector calculation

Chasles' relation $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$, vector addition and multiplication by a real number extend to space. Two non-zero vectors are collinear if one is a multiple of the other. The zero vector is collinear with every vector. Three vectors are coplanar if they belong to the same vector plane. If $\vec{u}, \vec{v}$ are not collinear, $\vec{u}, \vec{v}, \vec{w}$ are coplanar if and only if $\vec{w} = \alpha\vec{u} + \beta\vec{v}$ for some reals α, β.

Line and plane

The line through A with non-zero direction vector $\vec{u}$ is the set of points M such that $\overrightarrow{AM} = t\vec{u}$ for a real number t. The plane through A and spanned by two non-collinear vectors $\vec{u}, \vec{v}$ is the set of points M such that $\overrightarrow{AM} = s\vec{u} + t\vec{v}$ for real numbers s, t.

Three non-collinear points define a plane. Four points A, B, C, D, with A, B, C non-collinear, are coplanar if and only if $\overrightarrow{AD}$ is a linear combination of $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

Relative positions

Two lines are coincident, strictly parallel, intersecting or non-coplanar. Vectors collinear direction vectors characterise parallel lines in the broad sense. Non-collinear direction vectors do not guarantee intersection. A line and a plane: line contained in the plane, parallel and disjoint, or intersecting at a single point. Two planes: coincident, strictly parallel, or intersecting along a line.

Example in a cube. In ABCDEFGH , the faces $ABCD$ and $EFGH$ are parallel and the edges AE, BF, CG, DH correspond. The lines (AB) and (CG) are neither parallel nor intersecting: they are not coplanar.

Basis and coordinates

A basis of space is a family of three non-coplanar vectors. Every vector can be written uniquely $x\vec{\imath} + y\vec{\jmath} + z\vec{k}$. A coordinate system consists of an origin O and a basis.

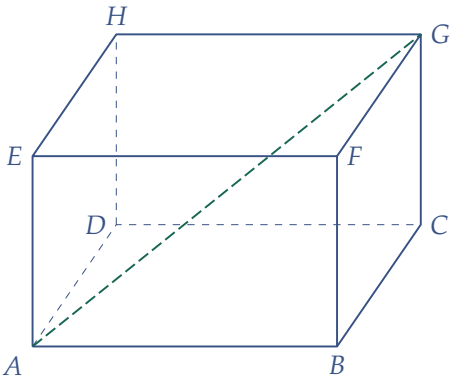
$$\overrightarrow{AB} = (x_B - x_A, y_B - y_A, z_B - z_A), \qquad I = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}, \frac{z_A + z_B}{2}\right).$$

In an orthonormal coordinate system, $\|\vec{u}\| = \sqrt{x^2 + y^2 + z^2}$ & $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$.

Example. In an orthonormal coordinate system, for $A(1,0,2)$, $B(3,-1,4)$, $\overrightarrow{AB} = (2,-1,2)$, $AB = 3$ and the midpoint is $(2, -\frac{1}{2}, 3)$.

Test a linear combination

Write the three coordinate equalities. Determine the coefficients from two equations when possible, then check the third. A contradiction proves non-membership; we do not delete the awkward equation.



$[AG]$ is a space diagonal, drawn in green dotted lines.
The hidden edges are dotted blue; D and F are not on (AG).

Build a reliable representation

10.1 Combinations and arbitrary bases

First use: S3 and S6.

Definition · Decomposing a vector

A basis of space is a triple of non-coplanar vectors. Every vector can be written uniquely as a combination of these three vectors. A basis is not necessarily orthogonal or made up of unit vectors.

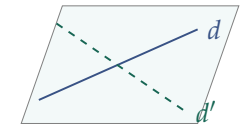
Worked example · In a tetrahedron

In a non-degenerate tetrahedron $OABC$, let $\vec{u} = \overrightarrow{OA}$, $\vec{v} = \overrightarrow{OB}$, $\vec{w} = \overrightarrow{OC}$. If I is the midpoint of $[AB]$, then $\overrightarrow{OI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v}$. Hence $\overrightarrow{CI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v} - \vec{w}$. These equalities use Chasles' relation and the midpoint, not lengths or angles.

A1 · Immediate application With the same basis, J is the midpoint of $[OC]$ and K satisfies $\overrightarrow{OK} = 2\vec{u} - \vec{v}$. Express $\overrightarrow{JK}$ then $\overrightarrow{IK}$.

10.2 Seeing is not enough to conclude

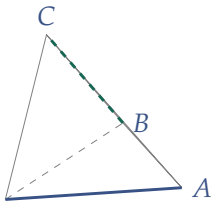
First use: S12.



intersecting, same plane



parallel, same plane



non-coplanar edges

Property · Relative positions of two lines

Two lines are either coplanar (intersecting or parallel) or non-coplanar. Vectors non-collinear direction vectors rule out parallelism, but do not prove an intersection.

Worked example · Two non-coplanar lines

In the tetrahedron $OABC$, the lines (OA) and (BC) are not coplanar. If a plane contained both, it would contain O, A, B, C, which would contradict the assumption that the tetrahedron is non-degenerate. An apparent intersection of their lines on the page would only be an effect of projection.

A2 · Read and argue On the three diagrams, name the relative position of the lines. Which geometric information accompanies the drawing to justify this answer? Give an example of lines that are not coplanar in a cube ABCDEFGH .

A3 · Refute "Two lines whose direction vectors are not collinear intersect." Disprove using the tetrahedron, then add a hypothesis that makes the statement true.

10.3 Characterising a plane by two directions

First use: S12 and S15.

Worked example · Membership and uniqueness

In a basis, let $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\overrightarrow{AM} = (2;3;5)$. We seek λ, μ such that $\overrightarrow{AM} = \lambda\vec{u} + \mu\vec{v}$. The first two coordinates require $\lambda = 2$, $\mu = 3$ and the third satisfies $5 = 2 + 3$. The point M lies in the plane through the point A with directions $\vec{u}, \vec{v}$. For $(2;3;4)$, the last equality fails: the point does not lie in it.

A4 · Partial guidance Are vectors $\vec{u} = (1;1;0)$ and $\vec{v} = (0;1;1)$ collinear? For which value of m is the vector $(2;m;3)$ a combination of $\vec{u}$ and $\vec{v}$?

A5 · Correct a mistake In an arbitrary basis, a pupil calculates the norm of the vector with coordinates $(1;1;0)$ as $\overline{2}$. Explain the missing assumption. Give an example with two unit vectors at an angle of 60 $^\circ$.

10.4 From the solid to the calculation

First use: S17.

Worked example · Midpoints and parallelism

If I and J are the midpoints of [OA] and [OB], $\overrightarrow{IJ} = \overrightarrow{IO} + \overrightarrow{OJ} = -\frac{1}{2}\overrightarrow{OA} + \frac{1}{2}\overrightarrow{OB} = \frac{1}{2}\overrightarrow{AB}$. The lines (IJ) and (AB) therefore have collinear directions. This calculation holds in any non-degenerate triangle.

A6 · Choice of strategy In the tetrahedron $OABC$, I is the midpoint of $[OA]$, J that of $[OB]$ and K that of $[OC]$. Prove that the planes (IJK) and (ABC) are parallel, then that O does not belong to (IJK). Do not use a Cartesian equation.

A first step: Compare $\overline{IJ}$ to $\overrightarrow{AB}$ and $\overrightarrow{IK}$ to $\overrightarrow{AC}$. For membership, decompose in $(\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC})$.

One minute, spoken aloud. Describe the tetrahedron and the three midpoints without showing the figure. Your partner must be able to place them. Specify the segments rather than "the point on the left".

Delayed review · S51 then S70. The same plane will be described by a normal vector, then an equation. The vector definition remains the way to check the calculation.

Investigation

Three midpoints in a cube - 40 min.

Consider the cube ABCDEFGH shown above. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are the respective midpoints of $[AB]$, $[AD]$ and $[AE]$. Let O the centre of the cube.

1. Express $\overline{IJ}$ and $\overrightarrow{IK}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$.

2. Prove that the points I, J and K are not collinear.

3. Determine whether O lies in the plane (IJK). Justify the answer by checking whether $\overrightarrow{IO}$ is a linear combination of $\overline{IJ}$ and $\overrightarrow{IK}$.

4. Why does reading the figure alone not allow us to prove either membership or non-membership of O in plane (IJK) ?